

Exercise Sheet #4: Electric Field

Discussion: Nov 12, 2025

Exercise 1: Discrete Charge Distribution

Two point charges of $+3.0 \mu\text{C}$ each are located at the points $(x, y) = (0, 2.0 \text{ m})$ and $(x, y) = (0, -2.0 \text{ m})$. Two additional point charges, each with charge q , are located at $(x, y) = (4.0 \text{ m}, 2.0 \text{ m})$ and $(x, y) = (4.0 \text{ m}, -2.0 \text{ m})$. The electric field at the point $(x, y) = (0, 0)$ due to the four charges is $\vec{E} = (4.0 \cdot 10^3 \hat{x}) \text{ N/C}$. Determine the value of q .

Solution

The electric field at the origin due to a point charge Q at position (x, y) is given by

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r},$$

where $\hat{r}$ is the unit vector pointing from the charge to the field point.

For the two charges at $(0, 2 \text{ m})$ and $(0, -2 \text{ m})$, the electric field components at the origin point along the y -axis. Since they are equal in magnitude and opposite in direction, they cancel each other:

$$E_y = 0 \quad (\text{net field from the two } 3.0 \mu\text{C} \text{ charges})$$

The distance to the origin of each q charge is

$$r = \sqrt{(4.0)^2 + (2.0)^2} \text{ m} = \sqrt{20} \text{ m},$$

and the electric field at the origin due to one q charge has magnitude

$$E_q = \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2}.$$

At the origin, the y -components of the fields from the charges at $(4 \text{ m}, 2 \text{ m})$ and $(4 \text{ m}, -2 \text{ m})$ cancel, while the x -components add.

Thus, the total field at the origin due to both q charges is

$$|E_x| = 2 \cdot E_q \cos \theta,$$

where $\cos \theta = \frac{x}{r} = \frac{4}{4.472} = 0.894$.

Given that the total field at the origin is

$$E_x = 4.0 \cdot 10^3 \text{ N/C},$$

we have

$$q = -\frac{E_x \cdot 4\pi\epsilon_0 \cdot r^2}{2 \cdot \cos \theta} = -\frac{4.0 \cdot 10^3 \cdot 2\pi \cdot 8.854 \cdot 10^{-12} \cdot 20}{0.894} \text{ C} = -5.0 \mu\text{C}.$$

Exercise 2: ...

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Solution

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